

EE P 533

Analog Circuits for Sensor Systems

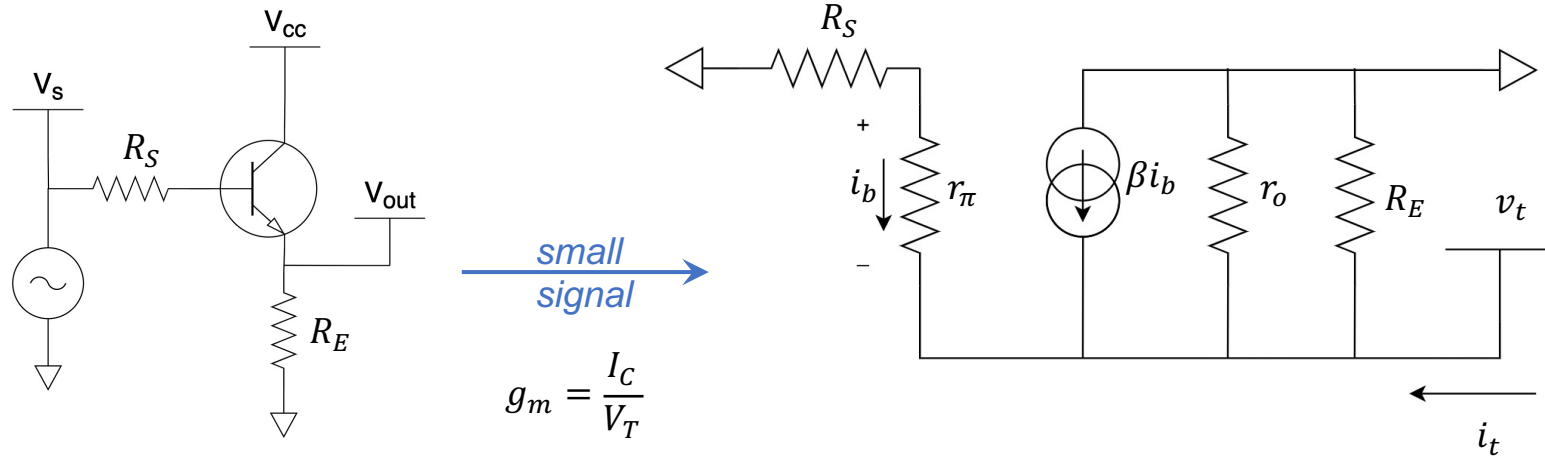
Spring 2026

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Review

- Derivation of output resistance, R_{out} , for BJT emitter follower amplifier (common collector)
- DC collector current bias, I_C , formulae

Output Resistance of Emitter Follower



By def: $R_{out} \equiv \frac{V_t}{i_t}$

By KCL: $i_t + i_b + \beta i_b = \frac{V_t}{R_E \parallel r_o}$

By KVL: $i_b = \frac{-V_t}{R_S + r_\pi}$

$i_t + \beta i_b \cong \frac{V_t}{R_E \parallel r_o}$

$i_t = V_t \left[\frac{1}{R_E \parallel r_o} + \frac{\beta}{R_S + r_\pi} \right]$

$\frac{V_t}{i_t} = \frac{R_S + r_\pi}{\beta}$

$R_{out} = \frac{V_t}{i_t} = \frac{R_S}{\beta} + \frac{1}{g_m} \cong \frac{1}{g_m}$

$\beta \gg 1$

$i_b = \dots$

$\frac{1}{R_E \parallel r_o} \ll \frac{\beta}{R_S + r_\pi}$

$r_\pi = \frac{\beta}{g_m}$ if $\frac{R_S}{\beta} \ll \frac{1}{g_m}$

BJT Collector Current

- Collector current depends on V_{BE} , and V_{CE} if including Early effect

$$I_C = I_S (e^{V_{BE}/V_T} - 1) \left(1 + \frac{V_{CE}}{V_A} \right)$$

- Excluding Early effect, e.g. $V_{CE}/V_A = 0.05$

$$I_C \cong I_S (e^{V_{BE}/V_T} - 1)$$

- When BJT is biased, $V_{BE} \cong 0.6V$
 $e^{V_{BE}/V_T} = 10^{10} \gg 1$

$$I_C \cong I_S e^{V_{BE}/V_T}$$

Week 4

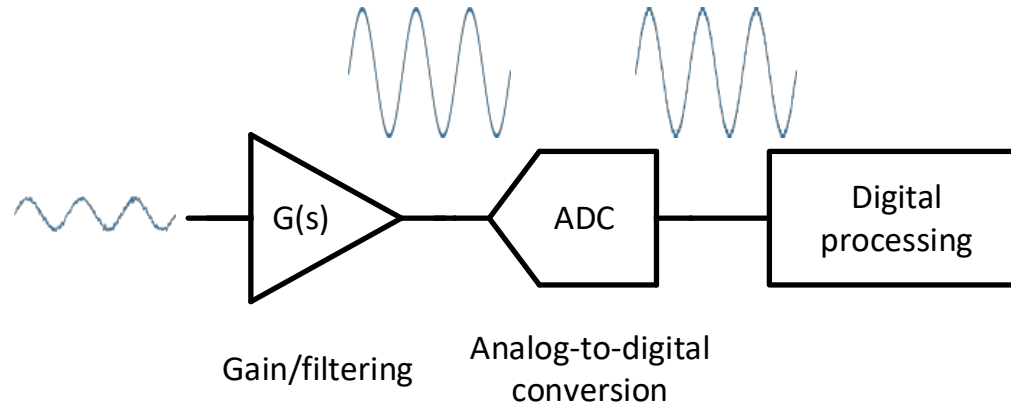
- Review of BJT
 - Derivation of output resistance, r_{out} , for BJT emitter follower
 - DC collector current bias, I_C , formulae
- AoE Section 2.5 – Negative Feedback
- AoE Chapter 4 – Operational Amplifiers

Overview

- Last time...
 - Small-signal BJT model
 - Current sources
 - Differential amplifiers
 - Field effect transistors
 - Small-signal model of a MOSFET
- Today...
 - Basic opamp behavior
 - Feedback
 - Frequency response
 - Non-ideal behavior

Lecture 4 – Opamps

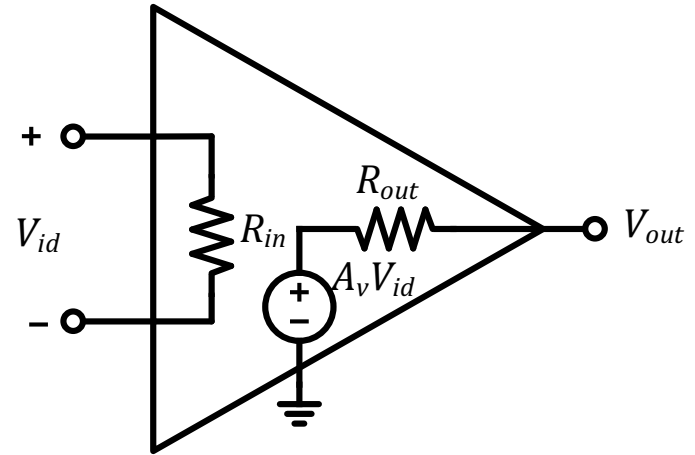
Analog Signal Conditioning



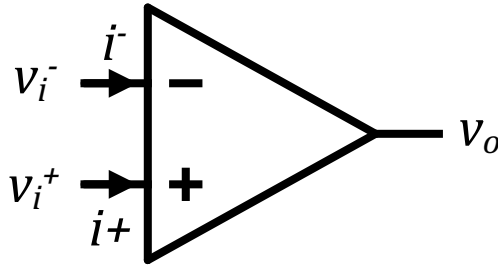
- The goal of signal conditioning is to prepare analog signals for analog-to-digital conversion while adding minimal noise
- Gain is used ensure that signals are optimized for ADC resolution and range, while filtering is required to ensure an acceptable signal-to-noise ratio (SNR)
- All components should be designed to minimize loading between stages

Ideal Opamp Characteristics

- Input impedance = ∞
- Output impedance = 0
- Voltage gain = ∞
- Common-mode rejection = ∞
- Offset voltage = 0
- No added noise
- Output can change instantaneously



Opamp “Golden Rules”

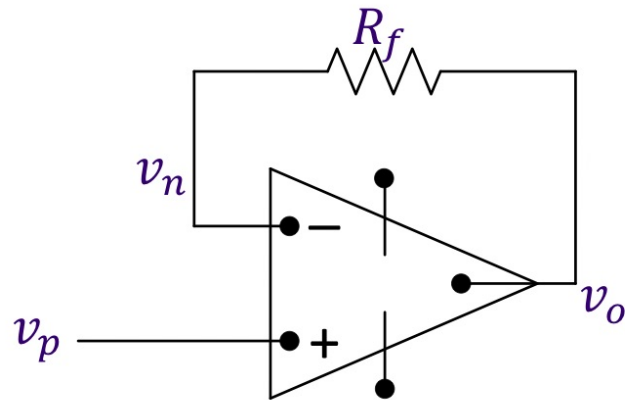


$$v_o = A_v(v_i^+ - v_i^-)$$

$$i_+ = i_- = 0$$

1. The output tries to do whatever it takes to make the voltage difference between the inputs zero
2. The inputs draw no current

- Negative feedback can be used to ensure operation in the linear region.



- If $(v_p - v_n)$ is too positive, then v_o increases, and thus v_n is increased by feedback loop.
- If $(v_p - v_n)$ is too negative, then v_o decreases, and thus v_n is reduced by feedback loop.

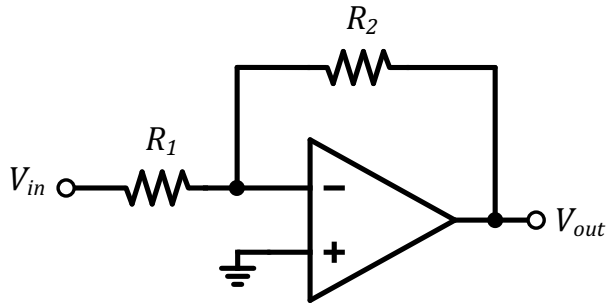
Opamp Design Caveats

1. Inputs and outputs should be within their prescribed ranges
 - Voltages outside these ranges can damage the opamp
2. Feedback should *always*¹ be negative and shouldn't introduce excessive phase shift
 - Phase lag in the feedback path can lead to oscillation!
3. Feedback is required at DC to set the operating point
 - This ties into 1, as the input voltage is otherwise ill-defined
4. Supply voltages should be 'bypassed' with capacitors to avoid instabilities due to supply rail wiring inductance
 - Most important for opamps with wide bandwidths

¹Electronic oscillators use positive feedback to compensate for energy losses in resonant structures, but this is outside the realm of linear circuit design.

Inverting/Non-inverting Amps

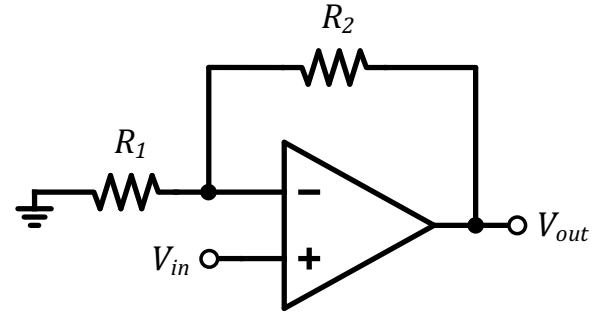
Inverting Amplifier



$$\frac{V_{in}}{R_1} = -\frac{V_{out}}{R_2}$$

$$\frac{V_{out}}{V_{in}} = -\frac{R_2}{R_1}$$

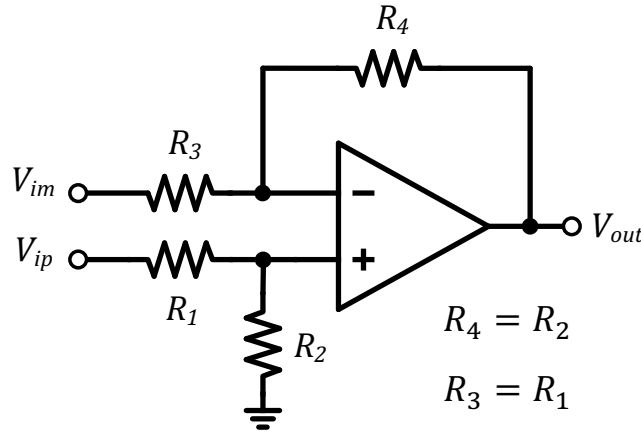
Non-inverting Amplifier



$$\frac{V_{in}}{R_1} = \frac{V_{out} - V_{in}}{R_2}$$

$$\frac{V_{out}}{V_{in}} = \left(1 + \frac{R_2}{R_1}\right)$$

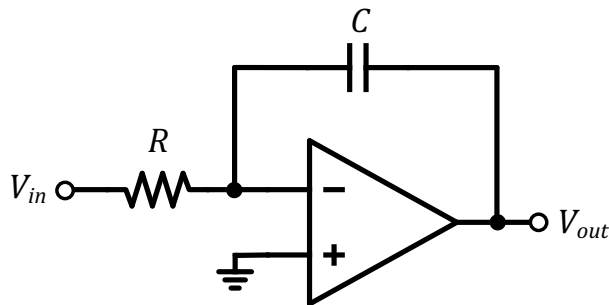
Difference Amplifier



$$\begin{aligned} V_{out} &= -\frac{R_2}{R_1} V_{im} + \frac{R_2}{R_1} V_{ip} \\ &= \frac{R_2}{R_1} (V_{ip} - V_{im}) \end{aligned}$$

- R_1 and R_2 form a divider which sets the non-inverting terminal voltage
- V_{ip} sees an inverting gain, while V_{im} sees a non-inverting gain
- Output voltage can be determined by superposition
- Can be used to achieve gain or perform differential to single-ended conversion (or both)

Opamp Integrator

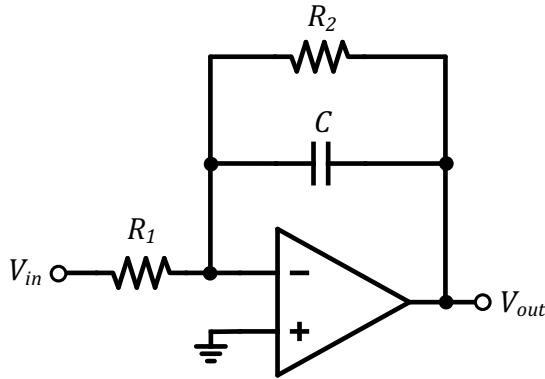


$$V_{out} = -\frac{Z_C}{R} V_{in} = -\frac{V_{in}}{sRC}$$

$$\Delta V_{out} = -\frac{V_{in}}{RC} \Delta t$$

- Opamps allow the design of *nearly* perfect integrators
- V_{in}/R is a constant current that is integrated onto C
- Avoids the restriction of $V_{out} \ll V_{in}$ for RC “integrators”
- Requires feedback at DC to have stable operating point!

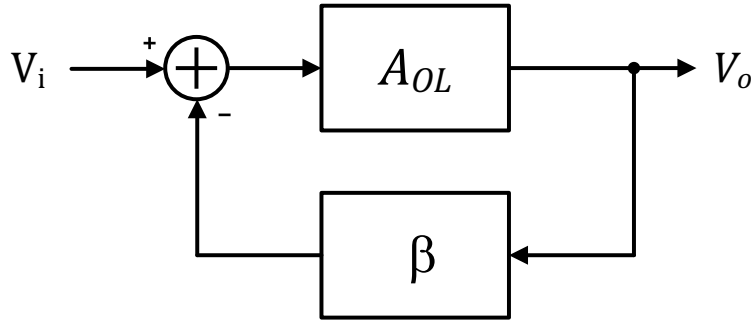
“Leaky” Integrator



$$\frac{V_{out}}{V_{in}} = - \frac{R_2 || \frac{1}{sC}}{R_1} = \frac{-R_2/R_1}{sCR_2 + 1}$$

- R_2 (or a reset switch, closed periodically) required to provide DC feedback
- At DC, the capacitor looks like an open circuit, so we just have an inverting amplifier
- At high frequency, the capacitor impedance dominates R_2 , so the structure behaves like an integrator (with some leakage current through R_2)

Feedback

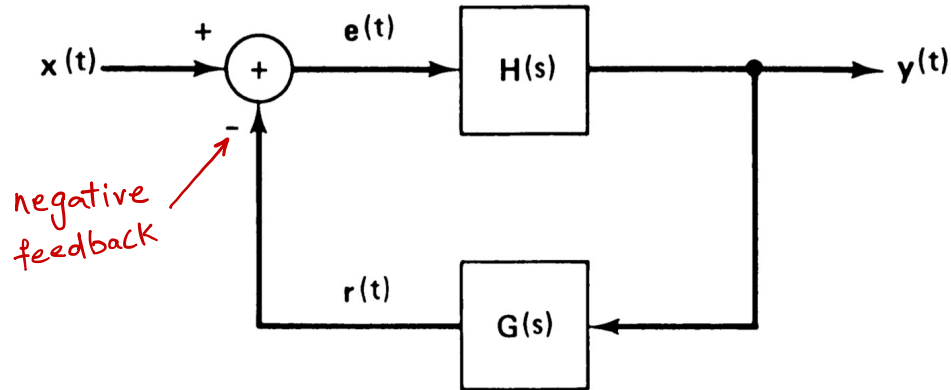


$$A_{CL} = \frac{V_o}{V_i} = \frac{A_{OL}}{1 + \beta A_{OL}}$$

$$\text{as } A_{OL} \rightarrow \infty, \quad A_{CL} \rightarrow \frac{1}{\beta}$$

- We use *negative* feedback to process the error $V_i - \beta V_o$, and A_{OL} should be “large” to minimize this error
- The combination of negative feedback and open-loop gain that is *high enough* desensitizes the transfer function to variations in open-loop gain
- The *feedback factor* β is the fraction of the output signal that is fed back to the input

How to analyze system shown below



$$e(t) = x(t) - y(t)$$

$$\Rightarrow E(s) = X(s) - R(s) \quad \text{--- (1)}$$

$$Y(s) = E(s) \cdot H(s) \quad \text{--- (2)}$$

$$R(s) = Y(s) \cdot G(s) \quad \text{--- (3)}$$

$$Q(s) = \frac{Y(s)}{X(s)} = \frac{H(s)}{1 + G(s)H(s)}$$

Substitute (2), (3) in (1).

$$\Rightarrow \frac{Y(s)}{H(s)} = X(s) - Y(s) \cdot G(s)$$

$$\text{Then, } Y(s) \left[G(s) + \frac{1}{H(s)} \right] = X(s) \quad \Rightarrow$$

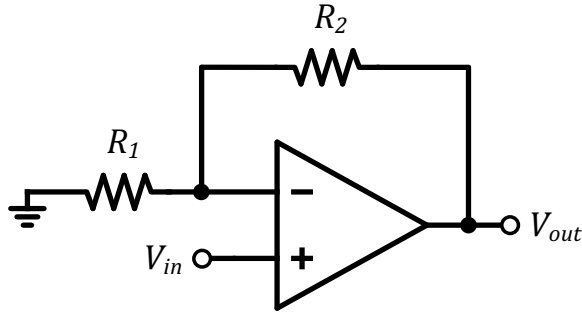
$$Q(s) = \frac{Y(s)}{X(s)} = \frac{H(s)}{1 + G(s) \cdot H(s)}$$

Closed loop transfer function.

Open loop transfer function

$$\Rightarrow \text{No feedback} \Rightarrow \frac{Y(s)}{X(s)} = H(s)$$

Non-inverting Amplifier



$$v_i^- = \frac{R_1}{R_1 + R_2} V_{out} \quad \rightarrow \quad \beta = \frac{R_1}{R_1 + R_2}$$

$$A_{CL} = \frac{A_{OL}}{1 + \beta A_{OL}} \approx \frac{1}{\beta} = 1 + \frac{R_2}{R_1}$$

- The output voltage, scaled by β , is fed back to the inverting terminal of the opamp
- The opamp *only processes the error voltage*, $v_i^+ - v_i^-$
 - This gives credence to our small-signal assumption when analyzing BJT/MOS circuits

Open-loop Frequency Response

- For internally-compensated opamps, the frequency response is approximately first-order (one pole) within the opamp's bandwidth
- The opamp bandwidth is its unity-gain, or transit, frequency ω_T
- The open-loop (DC) gain and bandwidth are related by the open-loop pole frequency:

$$\omega_T \approx A_0 \cdot \omega_{3dB}$$

Transfer function:

$$A_{OL}(j\omega) = \frac{A_0}{1 + j\omega\tau}$$

Magnitude:

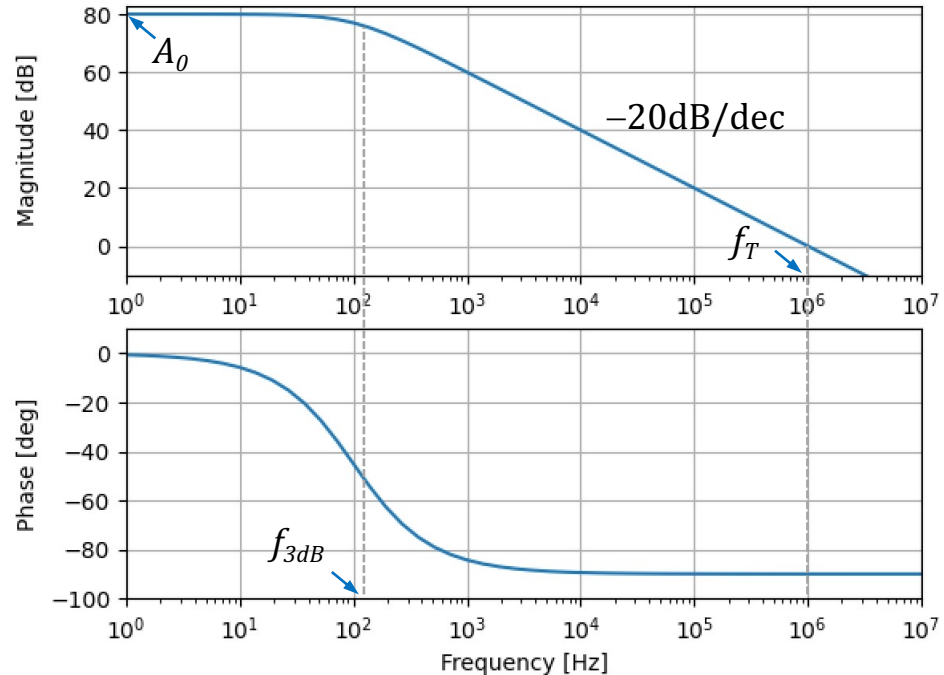
$$|A_{OL}(j\omega)| = \frac{A_0}{\sqrt{1 + \omega^2/\omega_{3dB}^2}}$$

Phase:

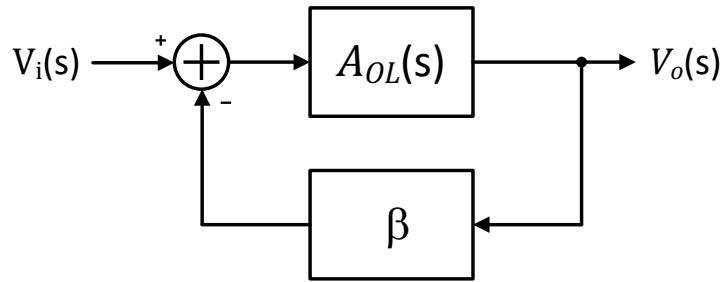
$$\angle A_{OL}(j\omega) = -\tan^{-1} \omega\tau$$

Open-loop Frequency Response

- As with any first-order system, the magnitude rolls off at 20dB per decade
- The single in-band pole results in 45° of phase lag at the pole frequency, f_{3dB}
- Beyond the unity-gain frequency (f_T), “non-dominant” poles create additional phase lag and steeper roll-off (not shown)



Closed-loop Frequency Response



$$A_{OL}(s) = \frac{A_0}{1 + s\tau}$$

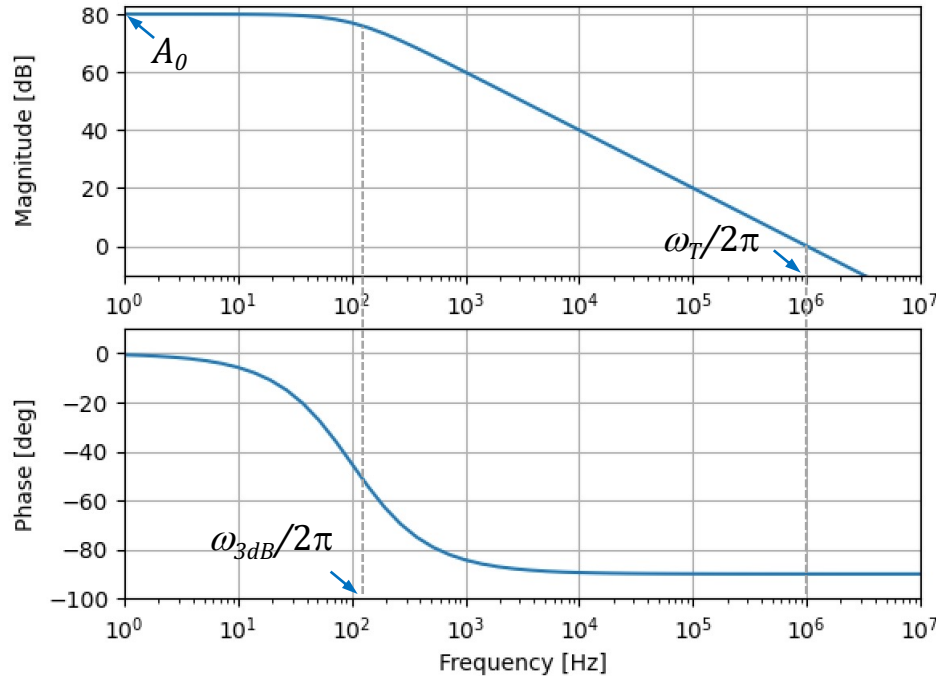
$$f_{3dB,OL} = \frac{f_T}{A_0}$$

$$A_{CL}(s) = \frac{A_0}{1 + s\tau + \beta A_0}$$

$$f_{3dB,CL} = \frac{1 + \beta A_0}{\tau}$$

- Opamps exhibit a constant *gain-bandwidth product*, approximately equal to f_T
- As the loop βA_0 increases, the 3dB bandwidth increases proportionally

Gain-Bandwidth Product



$$|A_{OL}(j\omega)| = \frac{A_0}{\sqrt{1 + \omega_T^2/\omega_{3dB}^2}} = 1 \quad (1)$$

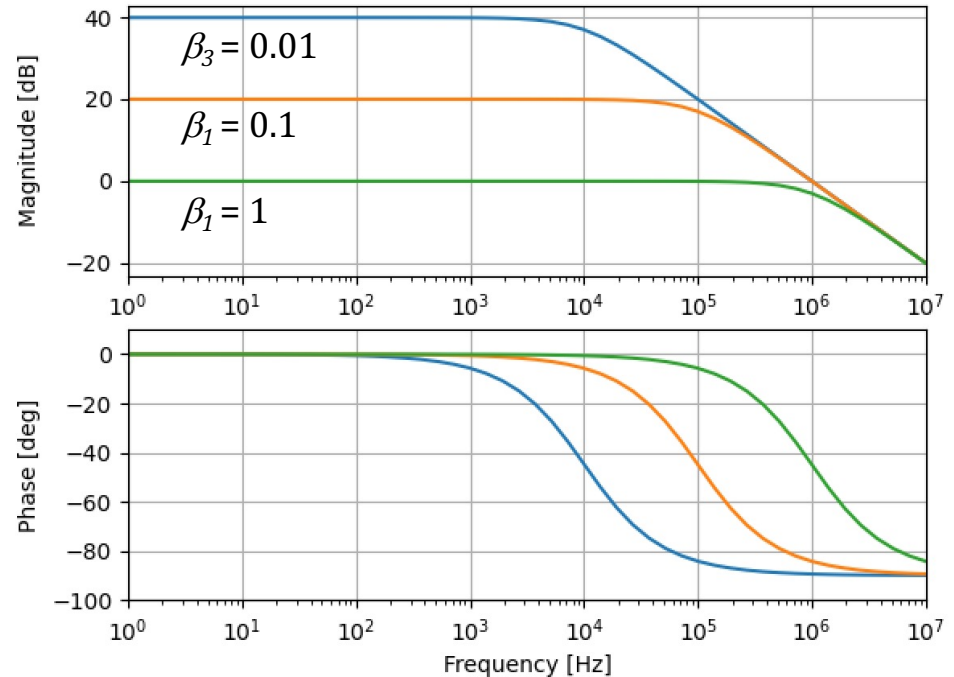
$$A_0^2 = 1 + \omega_T^2/\omega_{3dB}^2 \quad (2)$$

$$\omega_{3dB}^2 A_0^2 = \omega_{3dB}^2 + \omega_T^2 \quad (3)$$

$$\boxed{\omega_{3dB} A_0 \approx \omega_T} \quad (4)$$

Closed-loop Bandwidth

- Because $\omega_{3\text{dB}}$ and A_0 are constants, so is the gain-bandwidth product
- For different values of β , the magnitude response follows the same roll-off, but “breaks” at a different frequency
- An implicit assumption here is that the feedback network is frequency-independent (i.e. resistors only)



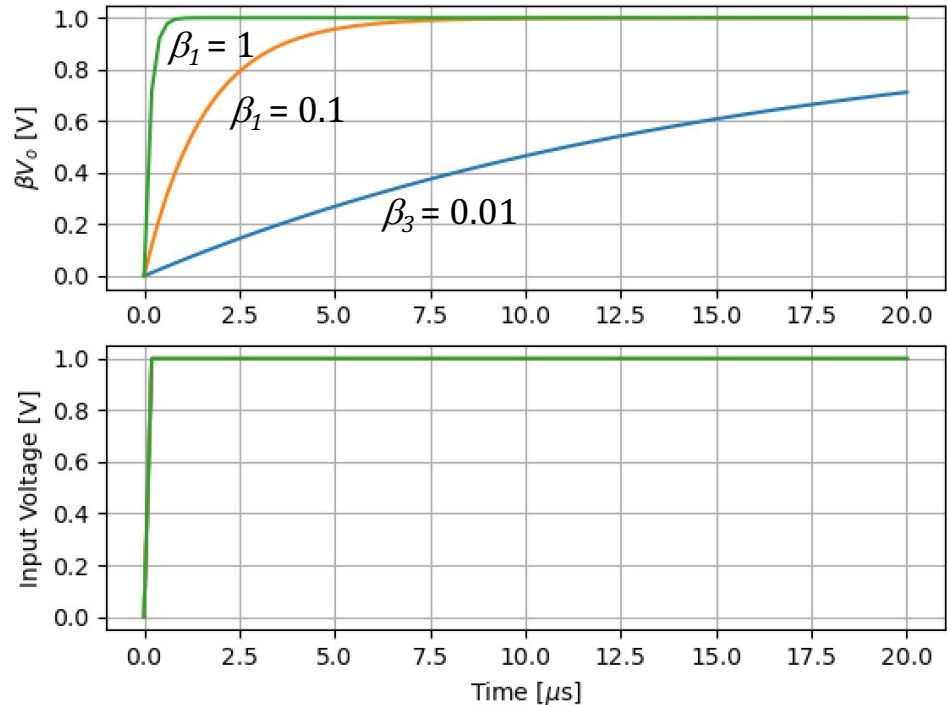
Closed-loop Settling Time

- As expected, settling time *decreases* with higher feedback factors

$$A_{CL}(s) = \frac{A_0}{1 + s\tau + \beta A_0}$$

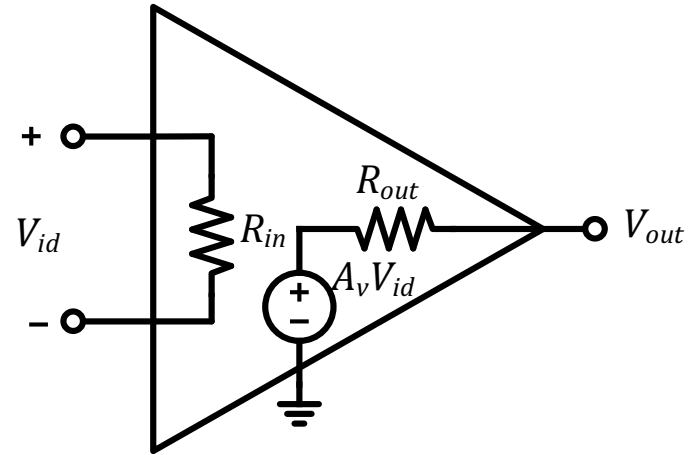
$$\tau_{CL} = \frac{\tau}{\beta A_0}$$

$$V_o(t) = u(t)(1 - e^{\frac{-t}{\tau_{CL}}})$$

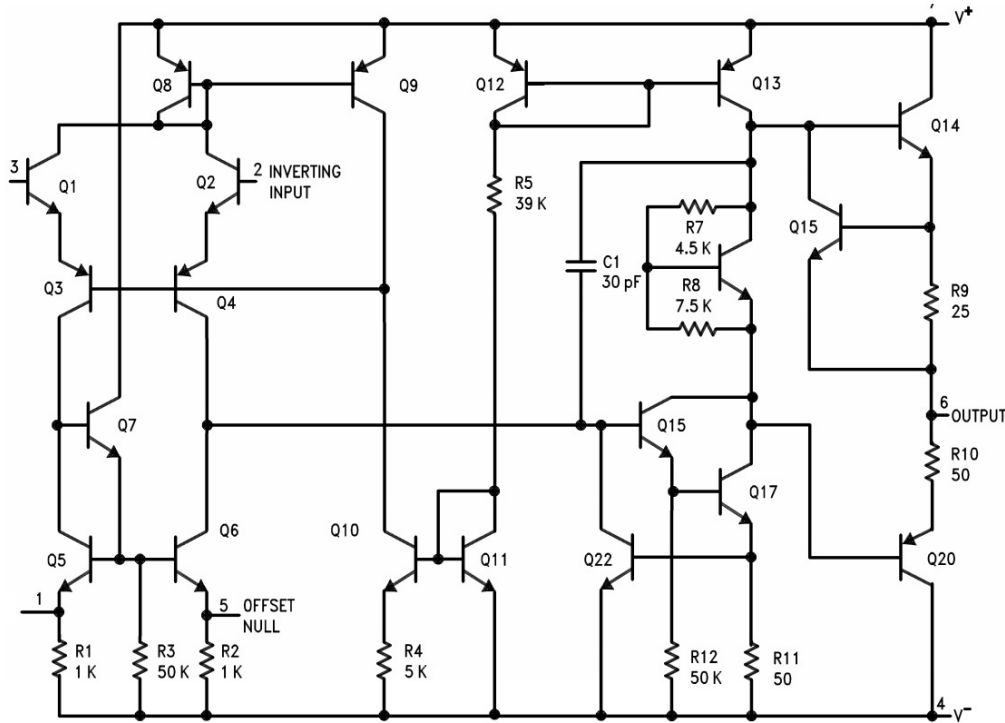


Real Opamp Characteristics

- Input impedance $1\text{M}\Omega - 1\text{T}\Omega$
- Output impedance: $1\Omega - 1\text{k}\Omega$
- Voltage gain $< 150\text{dB}$
- Common-mode rejection $< 120\text{dB}$
- Offset voltage: $1\mu\text{V} - 1\text{mV}$
- Nonzero input-referred noise
- Finite slew rate

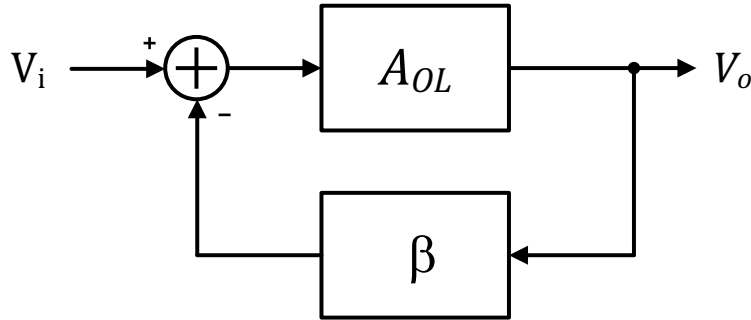


LM741 Architecture



- LM741 general-purpose opamp from Texas Instruments
- $Q_{1,2}$ form a common-emitter input pair
- C_1 is an internal compensation capacitance which sets the bandwidth
- $Q_{14,20}$ form a push-pull output stage

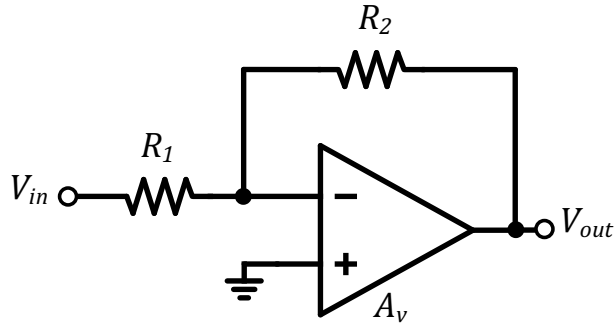
Finite Gain



A_{OL}	$\beta = 1$	$\beta = 0.1$
40 dB	1%	9%
60 dB	0.1%	1%
80 dB	0.01%	0.1%
100 dB	0.001%	0.01%

- Real opamps exhibit gain based on *finite physical parameters* (such as transconductance and output resistance)
- Finite open-loop gain results in an error in the output voltage

Finite Gain

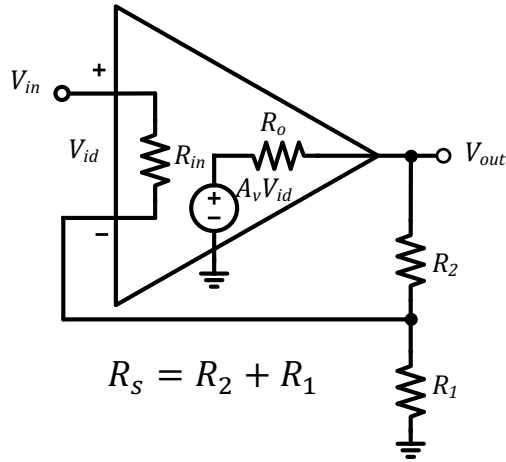


$$V_{out} = A_v(v^+ - v^-)$$

$$\frac{V_{out}}{V_{in}} = -\frac{\frac{R_2}{R_1}}{\frac{1}{A_v}\left(1 + \frac{R_2}{R_1}\right) + 1}$$

- Opamp inputs are only *exactly equal* if the open loop gain is *infinite*
- Finite *open-loop* gain of the opamp affects the precision of the *closed-loop* gain
- Even if R_1 and R_2 are *exactly* their expected values, there will be an error in the output voltage

Loading with Feedback



Without loading:

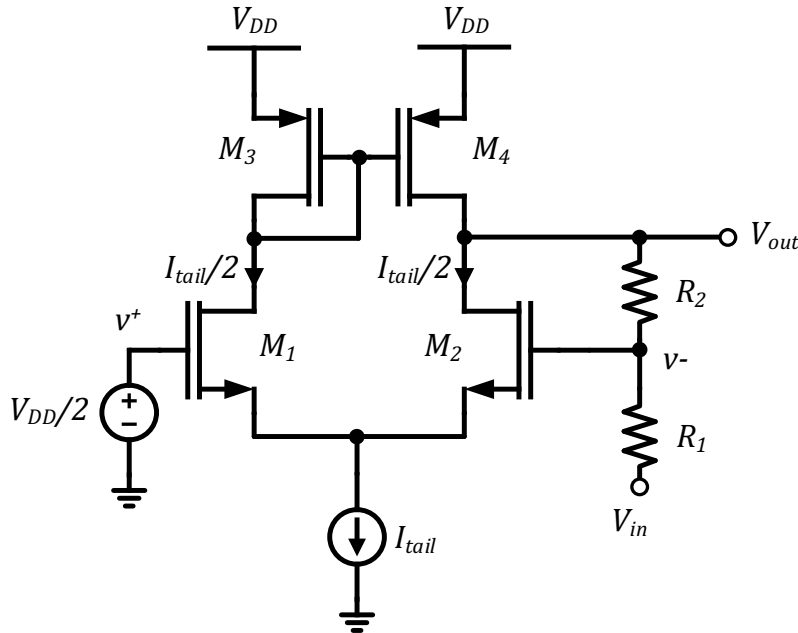
$$V_{out} = \frac{A_v}{1 + \beta A_v} V_{in}$$

With loading:

$$V_{out} = \frac{A_v}{\frac{R_o}{R_1 + R_2} + 1 + \beta A_v} V_{in}$$

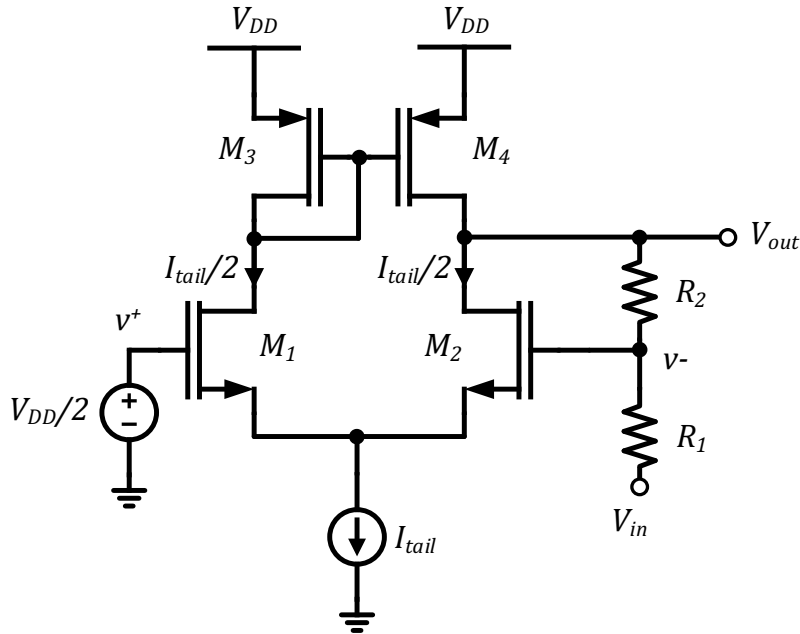
- The feedback network comprising R_2 and R_1 loads the opamp's output
- The effect of the loading depends on the magnitude of the opamp's output resistance relative to the feedback network resistance
- Loading due to the feedback network affects precision, but the effect is substantially alleviated by high open-loop gain

Input Mismatch



- Amplifier does “whatever it takes” to ensure $I_{D1} = I_{D2} = I_{bias}/2$
- Manufacturing variations result in a mismatch in the threshold voltages of M_1/M_2
- Because the source node is common to both transistors, their gate (or base, for BJT’s) voltages must differ
- Mismatch of other device parameters plays a role, but threshold mismatch dominates

Input Mismatch



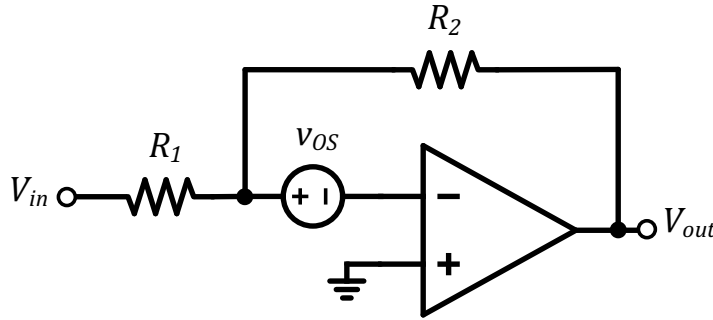
$$I_D = \kappa(V_{GS} - V_{th})^2$$

$$V_{GS1} = \sqrt{\frac{I_{tail}}{2\kappa}} + V_{th1} \quad V_{GS2} = \sqrt{\frac{I_{tail}}{2\kappa}} + V_{th2}$$

$$\Delta V_{th} = V_{th1} - V_{th2} = v_{os}$$

$$v^- = v^+ + v_{os}$$

Input Offset Voltage

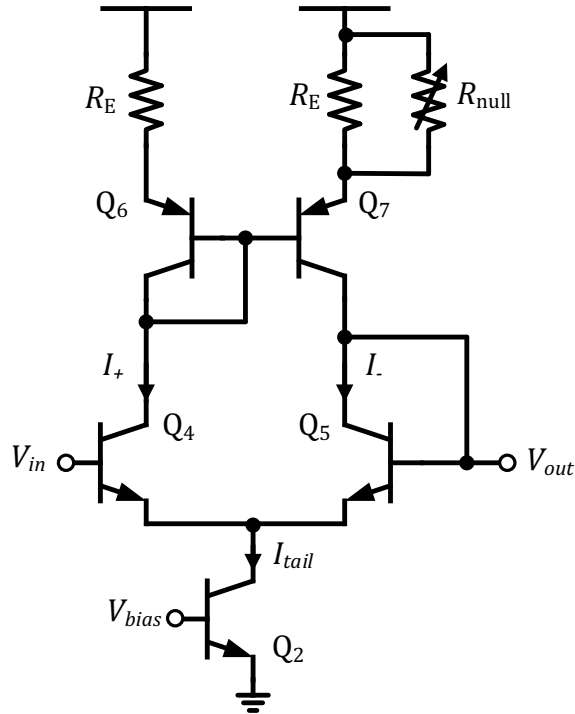


$$\frac{V_{in} - v_{os}}{R_1} = \frac{v_{os} - V_{out}}{R_2}$$

$$V_{out} = -\frac{R_2}{R_1}V_{in} + v_{os}\left(1 + \frac{R_2}{R_1}\right)$$

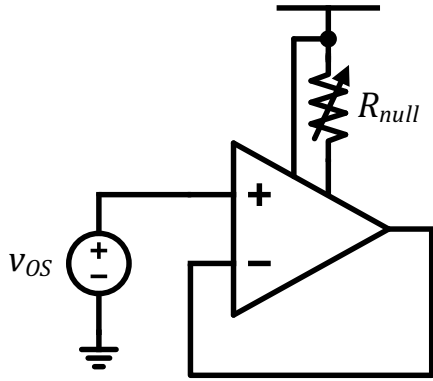
- Use superposition to include the effect of amplifier offset on the output voltage
- Input offset “sees” the gain of a non-inverting amplifier, while V_{in} is amplified as expected
- In general, offset voltage is much worse for FET-based opamps, primarily due to threshold voltage mismatch ([Ltpice: BJT OTA offset](#))

Offset Nulling



- I_+ and I_- nominally equal, and $V_{out} = V_{in} - V_{OS}$
- R_{null} increases the effective “size” of Q_7 in the current mirror, making I_- greater than I_+ ($I_+ + I_- = I_{tail}$)
- Since more current flows through Q_5 , its V_{BE} increases as R_{null} decreases
- R_{null} is adjusted until $V_{out} = V_{in}$
- [Offset nulling \(LTspice\)](#)

Offset Voltage Drift

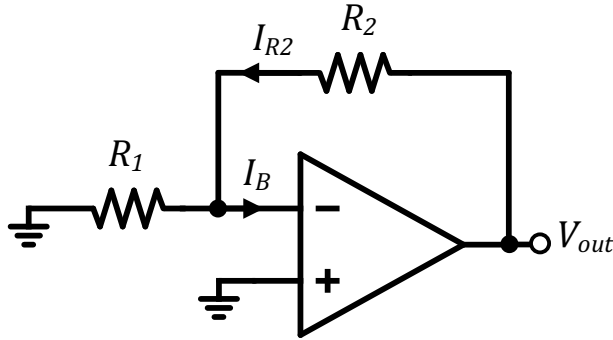


Before offset nulling: $V_{out} = v_{OS}(T)$

After offset nulling: $V_{out} = v_{OS}(T) - v_{OS}(T_0)$

- Dependence of offset voltage on temperature confounds simply “trimming” the offset at a given temperature as a blanket solution
- Nulling is performed at a single temperature (say, $T_0 = 25^\circ\text{C}$)
- However, an offset voltage drift of $10\mu\text{V}/^\circ\text{C}$ can cause output errors of a mV or more over typical operating ranges (e.g. -40 to 125°C)

Input Bias Current



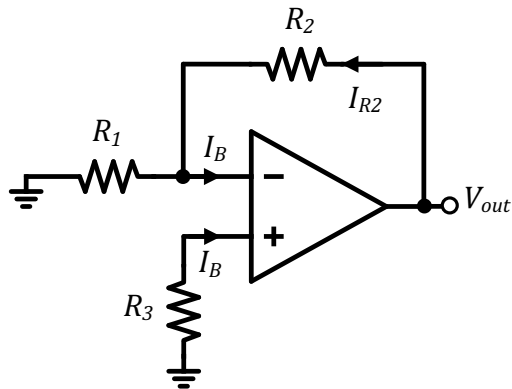
$$v_i^- = 0$$

$$I_{R2} = I_B$$

$$V_{out} = I_B R_2$$

- Opamp input currents cause an offset error in the output voltage proportional to the resistance of the feedback network
- Input currents can be as low as single-digit picoamperes (FET inputs), into the microamperes (BJT inputs)
- To avoid large voltage errors, use smaller resistances (but not so small as to load the opamp)

Bias Current Compensation

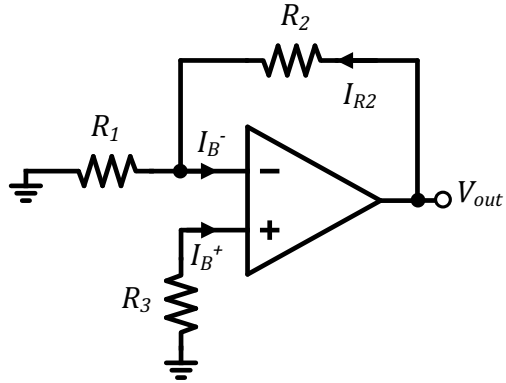


$$v_i^+ = -I_B R_3 = v_i^-$$

$$V_{out} = I_B R_2 - \left(1 + \frac{R_2}{R_1}\right) I_B R_3 = 0$$

- *Solution:* Add a resistor to the non-inverting terminal to equalize the voltage drops due to input bias current
- If the value of R_3 is equal to the *parallel combination* of R_1 and R_2 , the offset is cancelled
- It is good practice to keep R_1 and R_2 small, to minimize mismatch errors

Input Offset Current



$$I_{OS} = I_B^- - I_B^+$$

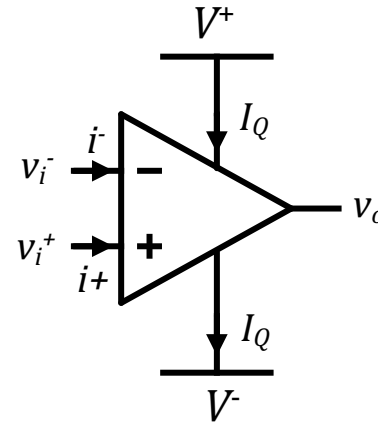
$$V_{out} = -\left(1 + \frac{R_2}{R_1}\right) I_B^+ R_3 + I_B^- R_2$$

$$V_{out} = R_2(I_B^- - I_B^+) = R_2 I_{OS}$$

- Input bias current is less problematic for balanced circuits
- However, a mismatch between I_B^+ and I_B^- (I_{OS}) will cause an error
- The only recourse is to use small resistance values in the feedback network (ensuring minimal loading of the opamp output)
- Input offset current is typically around 10 – 25% of input bias current

Opamp Supply Voltage

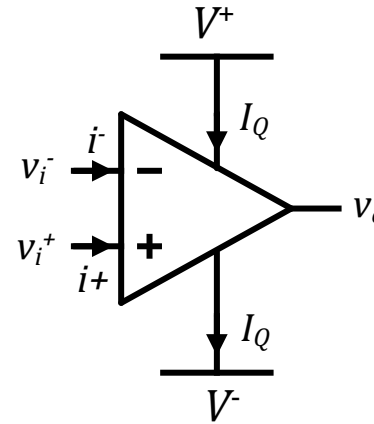
- Both bipolar and unipolar (ground-referenced) options abound
- Bipolar amplifiers can avoid AC-coupling for ground-referenced sensors
- Opamp power supply should be compatible with that of the ADC
- Most digital systems are unipolar, level conversion may be needed



$$V^- \leq v_i^+, v_i^-, v_o \leq V^+$$

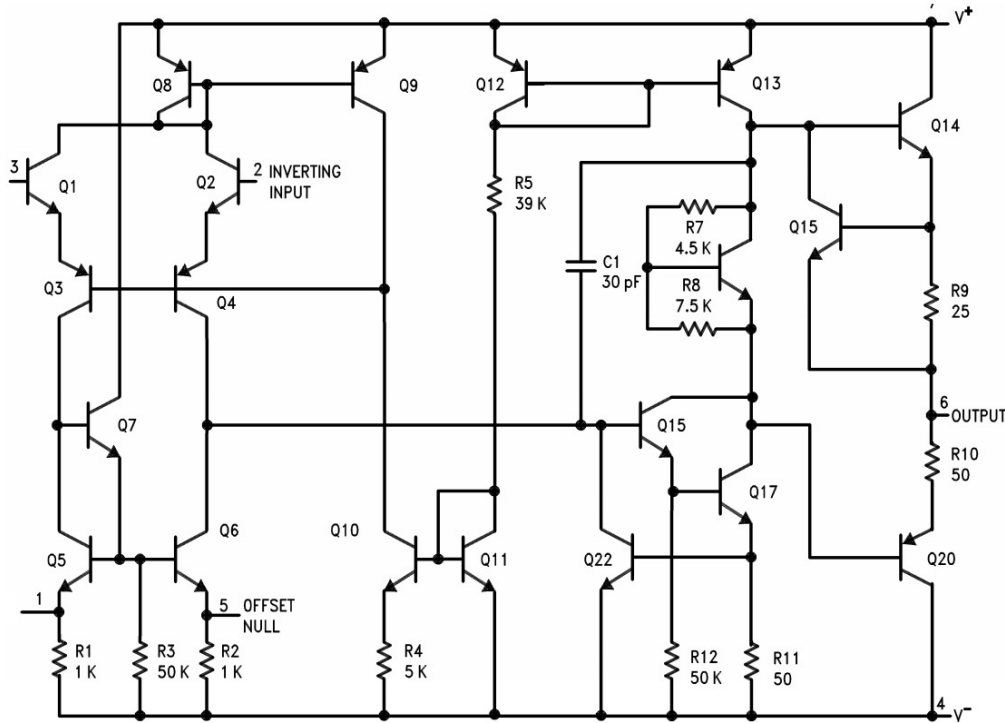
Opamp Quiescent Current

- Supply current opamps varies widely, dependent on application
- Both bandwidth and noise scale favorably with increasing supply current
- However, higher current means more heat generation
- For power-sensitive applications (e.g. energy harvesting or battery-powered), power efficiency is critical



$$f_T \propto I_Q \quad e_n \propto \frac{1}{\sqrt{I_Q}}$$

LM741 Specifications



Parameter	Typical
Open-loop gain	106 dB
Bandwidth	1.5 MHz
Input resistance	2 M Ω
Input offset voltage	1 mV
Offset voltage drift	15 μ V/ $^{\circ}$ C
Input bias current	80 nA
Input offset current	20 nA
Supply current	1.7 mA